# Arista Access Node MG-ACC-70 QoS Congestion Event SOP

**Node:** MG-ACC-70 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9400

## Description

This Standard Operating Procedure (SOP) is designed to address QoS congestion events on the Arista Access node MG-ACC-70 at the MG Road Core site. The failure domain of microbursts, interface tail drops, and QoS queue starvation is a critical issue that can impact network performance and service availability. The underlying protocol and hardware mechanics contributing to this failure domain include the Arista EOS (Extensible Operating System) QoS feature set, which relies on the TCAM (Ternary Content-Addressable Memory) for packet classification and policing. Additionally, the Arista access node's role in the network topology, which involves forwarding traffic from the edge to the core, makes it a critical component in the network's overall QoS strategy. The combination of these factors, along with the specific configuration and settings of the MG-ACC-70 node, can lead to congestion events when the network is under heavy load or experiencing microbursts of traffic. This SOP outlines the diagnostic and resolution steps required to identify and address the root cause of the QoS congestion event on the MG-ACC-70 node.

## Pre-requisite Checks

- Verify the MG Road Core site's environmental conditions are within normal operating parameters (temperature, humidity, power availability).

- Ensure the redundant link status on the MG-ACC-70 node is operational and not experiencing any issues.

- Dispatch remote hands to the MG Road Core site to perform any necessary physical checks or repairs.

## Potential Root Causes

| Cause | Probability |
|---|---|
| Misconfigured QoS policies or class maps | High |
| Insufficient TCAM resources for packet classification and policing | Medium |
| Interface congestion or hardware issues | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show qos config all, Expected Output: The output will display the current QoS configuration, including class maps, policy maps, and service policies., Description: This command will provide an overview of the QoS configuration on the MG-ACC-70 node, allowing the engineer to identify any misconfigured policies or class maps that may be contributing to the congestion event.
```

```
Command: show qos policers, Expected Output: The output will display the current policer configuration, including policer rates and burst sizes., Description: This command will provide information on the policer configuration, allowing the engineer to identify any issues with policer rates or burst sizes that may be contributing to the congestion event.

Command: show platform tcam utilization, Expected Output: The output will display the current TCAM utilization, including the percentage of TCAM resources allocated to each table., Description: This command will provide information on the TCAM utilization, allowing the engineer to identify any issues with TCAM resources that may be contributing to the congestion event.
```

## Resolution Steps

Step 1: Step: Update QoS policies and class maps, Command: configure qos policy-map update, Description: The engineer will need to update the QoS policies and class maps to ensure they are correctly configured and aligned with the network's QoS strategy.

Step 2: Step: Adjust policer rates and burst sizes, Command: configure qos policer update, Description: The engineer will need to adjust the policer rates and burst sizes to ensure they are correctly configured and aligned with the network's QoS strategy.

Step 3: Step: Allocate additional TCAM resources, Command: configure platform tcam allocation update, Description: The engineer will need to allocate additional TCAM resources to ensure the TCAM is properly allocated and can handle the traffic load.

## Post-Check Verification

```
Command: show qos config all, Expected Output: The output will display the updated QoS configuration, including class maps, policy maps, and service policies., Description: This command will verify that the QoS configuration has been updated correctly and is aligned with the network's QoS strategy.

Command: show qos policers, Expected Output: The output will display the updated policer configuration, including policer rates and burst sizes., Description: This command will verify that the policer configuration has been updated correctly and is aligned with the network's QoS strategy.

Command: show platform tcam utilization, Expected Output: The output will display the updated TCAM utilization, including the percentage of TCAM resources allocated to each table., Description: This command will verify that the TCAM utilization has been updated correctly and is aligned with the network's QoS strategy.
```

## Roll-back Procedures

- Step: Revert QoS policies and class maps, Command: configure qos policy-map revert, Description: The engineer will need to revert the QoS policies and class maps to their previous state if the fix fails to resolve the congestion event.

- Step: Revert policer rates and burst sizes, Command: configure qos policer revert, Description: The engineer will need to revert the policer rates and burst sizes to their previous state if the fix fails to resolve the congestion event.

- Step: Revert TCAM resources, Command: configure platform tcam allocation revert, Description: The engineer will need to revert the TCAM resources to their previous state if the fix fails to resolve the congestion event.